CS-1101
PROGRAMMING FUNDAMENTALS

Lab 05

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- Control Structures
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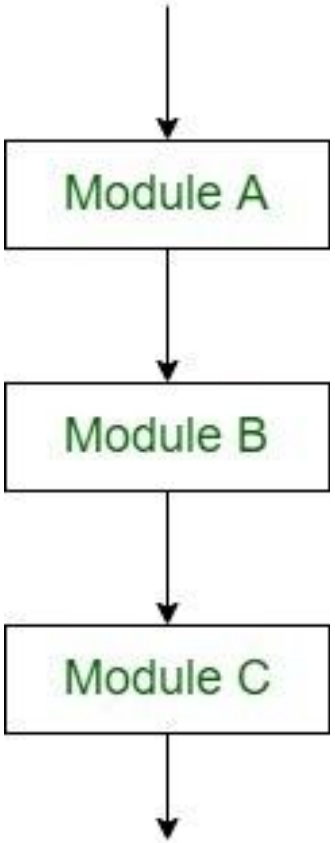
Lab Objectives

- Learn how to use simple if and else statements to control the flow of a program based on different conditions and inputs.

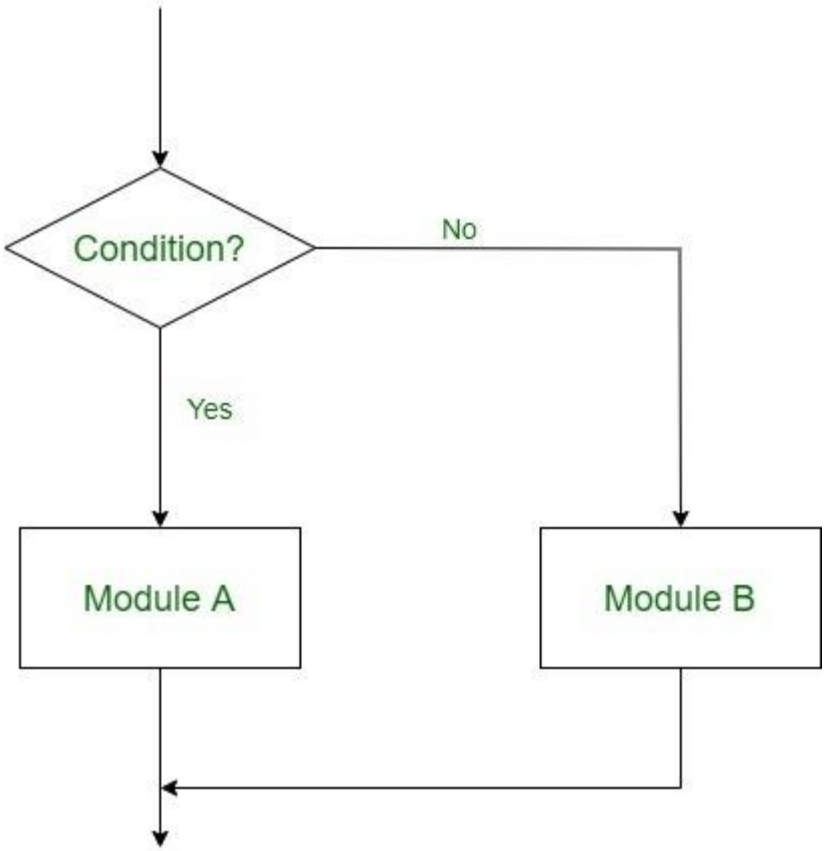
Control Structures

- Control Structures are just a way to specify flow of control in programs.
- There are three basic types of flow of control, known as:
  - Sequential flow
  - Conditional flow
  - Repetitive flow

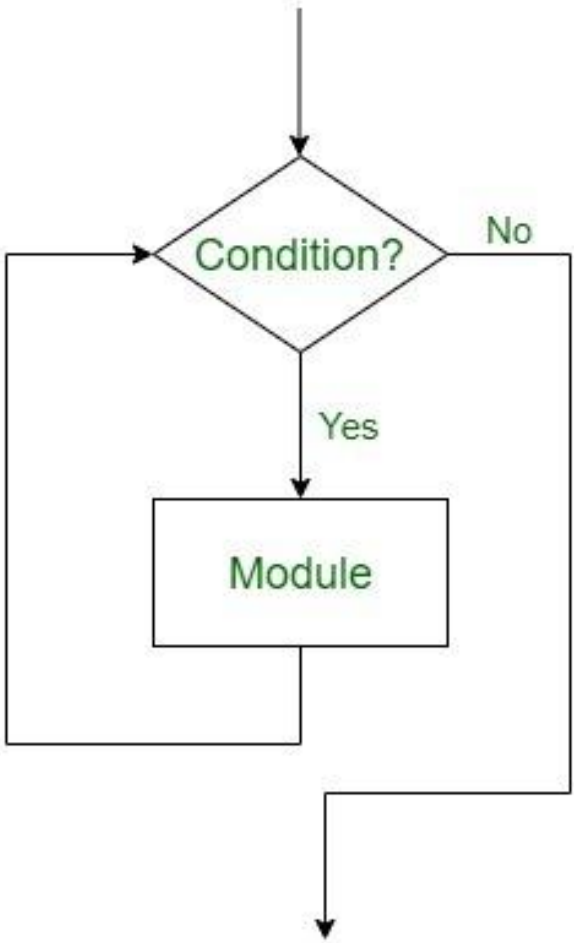
Sequential Flow



Conditional Flow

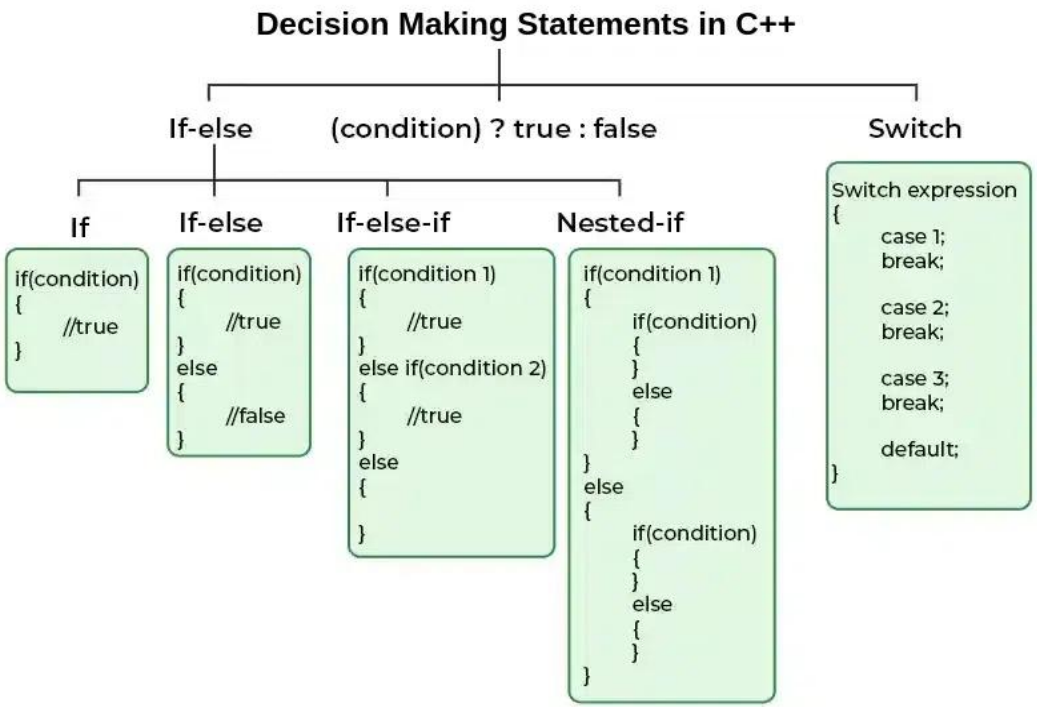


Repetitive Flow



Decision Control Structures – Conditional Flow

- Decision-making in C++ involves the usage of conditional statements (also called decision control statements) to execute specific blocks of code primarily based on given situations and their results.



if in C++

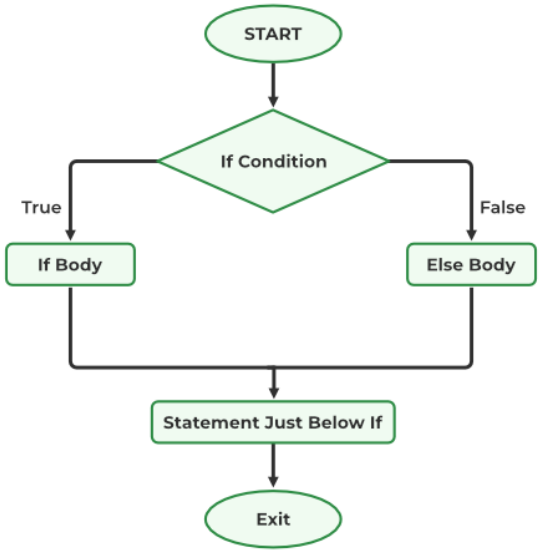
- if statement is the simplest decision-making statement.
- It allows the execution of a block of code if the given **condition is true**.
- The body of the 'if' statement is executed only if the given condition is true.

```
if (condition) {
// code to be executed if the condition is true
}
```

if-else in C++

- The if-else decision-making statement allows us to make a decision based on the evaluation of a given condition.
- If the given condition evaluates to true then the code inside the 'if' block is executed and in case the condition is false, the code inside the 'else' block is executed.

```
if (condition) {
// Code to be executed if the condition is true
}
else {
// Code to be executed if the condition is false
}
```



Basic Tasks – if

- Write a program that takes an integer as input and checks if it is a positive number. If it is positive, print "The number is positive." Use only the if statement.
- Write a program to determine if a given integer is even. Use only the if statement to display "The number is even" if the condition is met.
- Write a program to find the largest of three numbers entered by the user. Use only if statements to compare the numbers and print the largest one.
- Develop a program that takes a student's marks (out of 100) as input and determines their grade using the following scale:
  - 90 and above: A+
  - 80–89: A
  - 70–79: B
  - 60–69: C
  - Below 60: Fail

  Use only if statements for the conditions.

Basic Tasks - if

- Write a program that simulates a traffic signal system. The user enters the color of the signal as R (Red), G (Green), or Y (Yellow), and the program responds with:
  - Red: Stop
  - Green: Go
  - Yellow: Slow Down

  Use only if statements for this simulation.

Basic Tasks – if-else

- Write a program to simulate a basic ATM machine. The program should:
  - Ask the user for their account balance.
  - Allow the user to input the withdrawal amount.
  - Check if the withdrawal amount is less than or equal to the account balance.
  - If yes, deduct the amount from the balance and display the remaining balance.If no, display a message: "Insufficient balance.“
- Write a program to check if a person is eligible for a loan based on the following rules:
  - The person must have a minimum salary of $25,000.
  - If the person has more than 5 years of work experience, they are eligible even if their salary is $20,000.

Use if-else statements to determine the eligibility.

Challenge - Employee Payroll Management System

- You are tasked with designing a program for a company's payroll system. The system should calculate the net salary of an employee based on the following rules:

1. **Basic Salary and Allowances:**
   1. Employees are categorized into three grades: A, B, and C.
      1. **Grade A:** Basic salary = $50,000
      2. **Grade B:** Basic salary = $30,000
      3. **Grade C:** Basic salary = $20,000
2. **Bonuses:**
   1. Employees with years of service greater than or equal to 10 receive a **10% bonus** on the basic salary.
   2. Employees with years of service between 5 and 9 (inclusive) receive a **5% bonus**.
   3. Employees with less than 5 years of service receive **no bonus**.
3. **Tax Deductions:**
   1. A **20% tax** is deducted if the total gross salary (basic salary + bonus) exceeds $40,000.
   2. A **10% tax** is deducted if the gross salary is between $30,000 and $40,000 (inclusive).
   3. No tax is deducted for gross salaries below $30,000.
4. **Net Salary Calculation:**
   1. The net salary is calculated as:
      Net Salary = Gross Salary - Tax

USE ONLY if Statements

REFERENCES

- https://www.geeksforgeeks.org/cpp-decision-making/